

Hypothesis-Search Report: D2 Receptor-Arrestin-GSK-3β Tau Kinase Model in Alzheimer Disease

Hypothesis ID: `d2r_arrestin_gsk3b_tau_model` · **KB status:** EMERGING · **Search date:** 2026-09-21 **Run type:** Literature hypothesis-search (no dataset provided; no omics/structural analysis performed)

Executive Judgment

Verdict: WEAKLY / PARTIALLY SUPPORTED as assembled, with the central disease-relevant claim UNTESTED and the core directional premise contradicted by the dominant AD-dopamine literature.

The hypothesis is best described as a *mechanistically plausible but unverified assembly* of edges borrowed from three different domains:

1. **Upstream edges are individually well established** (high confidence). The noncanonical D2R → β-arrestin2/PP2A/Akt → GSK-3β pathway is a mature, independently replicated mechanism

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P 17681085

P 23188793

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P 23380502

and GSK-3β is repeatedly identified as the principal tau kinase
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P 23321003

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P 38160617

The load-bearing distinction — that aripiprazole, unlike haloperidol, preserves pAkt—

pGSK3 β under D2R hyperactivity — is real but was measured in a **schizophrenia/DISC1 cortical-neuron neurite model, not tau, not AD** (P 30597182).

2. **The disease-relevant terminal claim is untested** (the decisive gap). A PubMed search for **aripiprazole tau phosphorylation amyloid Alzheimer model** returned **zero** papers. No study has measured tau or GSK-3 β -dependent phospho-epitopes after aripiprazole in any AD/tauopathy model. The one in-vivo AD exposure (5xFAD, 1 mg/kg $\times$ 2 months, n=6/group) reports a **glutamatergic/behavioral** benefit and explicitly did not measure tau, with the authors themselves flagging a **safety** constraint (P 39174788).

3. **The core premise runs against the prevailing AD-dopamine model** (the strongest caveat). The seed posits D2R **over-activation** as pathogenic. The dominant experimental framework instead describes **early VTA dopaminergic degeneration / hypodopaminergia** in AD models, rescued by levodopa (P 28367951), (P 29778899), and human genetics associate **lower** D2 receptor function (DRD2 Taq1A A1) with worse memory and higher AD risk (P 28965318), (P 27647283).

A 2026 review couples dopamine **depletion** to GSK-3 β **activation** — the opposite direction (P 41930581).

Under these frameworks, any 5xFAD aripiprazole benefit is more parsimoniously explained by a partial agonist **restoring deficient dopaminergic tone**, not by blocking arrestin-mediated tau kinase activation.

Additional headwinds: direct GSK-3 β inhibition failed to help AD cognition in RCTs (tideglusib null;

(P 31156177), and tau hyperphosphorylation in AD is substantially explained by **PP2A phosphatase failure**

(APOE- ϵ 4-linked) that needs no D2R input

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P 22299660
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P 28720530

The seed's own use of PP2A is internally strained, since PP2A also dephosphorylates GSK-3 β -Ser9

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P 26484916
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Net: the hypothesis earns its EMERGING label and its own falsification test is feasible
 (UNC9994 exists and is arrestin-biased;

P 22025698
).

It should **not** be promoted to a curated causal edge until the tau arm has direct support and the directional conflict is resolved.

Evidence Matrix

PMID	Type	Stance	Mechanistic claim tested	Key finding	Context	Confidence limitation
30597182	in vitro	support (load-bearing)	aripiprazole preserves pAkt-pGSK3 β vs haloperidol under D2R hyperactivity	"Unlike haloperidol, aripiprazole prevented downregulation of phospho...Akt-pGSK3 β "	cortical neurons, DISC1/schizophrenia	Indirect: no readout, no AD
41192583	in vitro	support / qualify	aripiprazole arrestin ligand profile at D2/D3	"weak or no agonist but potent antagonist activity" at arrestin	recombinant human D2/D3	Indirect; recombinant
39174788	model organism	support (premise only)	chronic aripiprazole benefits an AD model	improved NORT; +7–8% glutamate PET; glutamatergic recovery	5xFAD, 1 mg/kg $\times$ 2 mo, n=6/grp	Small n; glutamate; tau; author caveat
34173272	human clinical	refute	antipsychotic effect on AD cognition	APs $\rightarrow$ faster cognitive decline & dementia progression; worse in APOE ϵ4	NILVAD, n=509 mild-moderate AD	Class-level; aripiprazole specific; confounding indication
17681085	in vitro / human	support (backbone)	DRD2 $\rightarrow$ AKT1/PP2A/ β arr2 $\rightarrow$ GSK-3 β	complex "dephosphorylates/inactivates AKT1 thereby activating GSK-3beta"	schizophrenia cells/brain	Backbone AD/tau
23188793	model organism	support (backbone)	β arr2/Akt/PP2A/GSK3 β in D2R neurons; atypicals antagonize D2R/ β arr2	D2GSK3 β -KO mimics antipsychotic action	mouse D2R neurons	Behavior n
23380502						Striatum n

PMID	Type	Stance	Mechanistic claim tested	Key finding	Context	Confidence limitation
	model organism	support (backbone)	D2R β arr2/PP2A/Akt complex dephosphorylates Akt	apomorphine recruits PP2A-C to Akt	mouse striatum	
23321003	in vitro / model	support (terminal)	GSK-3 β is principal tau kinase; GSK-3 β /PP2A balance	shifting balance lowers A β -induced tau	cortical neurons + Tg mouse	Supports GSK3 β →tau
22025698	in vitro / model	support (reagent)	UNC9994 identity & pharmacology	arrestin-biased D2R agonist from aripiprazole scaffold; effect β -arrestin2-dependent	recombinant + mice	Enables discrimination not AD/tau
22845053	in vitro	support (reagent)	UNC9994 structure-functional-selectivity	arrestin-biased D2R agonists from aripiprazole scaffold	recombinant	Reagent characterization
26129680	model organism	support (reagent)	UNC9994 in-vivo behavior	improves NORT in hypoglutamatergic mice	mice	Schizophrenia AD
28367951	model organism	competing	VTA DA neuron loss drives AD memory deficit	pre-plaque VTA DA loss lowers hippocampal/NAC DA	Tg2576	Opposes or activation (hypodopamine)
29778899	model organism	competing	DA loss impairs hippocampus-NAC; DA restoration rescues	levodopa rescues deficits	Tg2576	DA restoration beneficial
29630556	review / human	competing	VTA volume associates with AD cognition	human VTA–cognition link	prodromal AD	Orientation

PMID	Type	Stance	Mechanistic claim tested	Key finding	Context	Confidence / limitation
33684513	model organism	competing	c-Abl drives VTA DA degeneration	nilotinib prevents DA loss, lowers A β , restores memory	Tg2576	Alternative upstream of autophagy
31156177	human clinical	qualify / refute	GSK-3 β inhibition improves AD cognition	GSK-3 inhibitors ineffective; tideglusib null (SMD=-0.02, p=0.89)	5 RCTs, N=568	Weakens t node therapeutic rationale
22299660	review / human	competing	PP2A failure drives AD tau	multiple PP2A deficits in AD brain	human AD brain	Parallel, D independent driver
28720530	in vitro	competing	APOE downregulates PP2A	ApoE lowers PP2A	AD relevance	Links tau-phosphatase to APOE $\epsilon 4$
26484916	in vitro	qualify	PP2A dephosphorylates GSK-3 β -Ser9	PP2A suppresses GSK-3 β -Ser9	brain extracts / HEK293FT	Complicated one-way P
38350636	review	competing	amyloid + GSK3 β /CDK5 + PP2A	CDK5 is a parallel tau kinase	AD review	Multiple co-kinases
36760217	model organism	competing	A β 42 lowers pAkt/pGSK3 β , raises pTau	A β drives Akt/GSK3 β /tau axis	A β 42 icv mice	Amyloid upregulation of GSK3 β via D2R
41930581	review	contradict (direction)	dopamine DEPLETION activates GSK-3 β /tau	"dopamine depletion, leading to increased GSK-3 β activity"	2026 narrative review	Opposite direction t
28965318	human (genetic)	contradict (direction)	DRD2 function and AD risk	DRD2 Taq1A A1 (low D2 density) may magnify AD risk	aging African-Americans	Association of tau readout
27647283					n=525, age 60	

PMID	Type	Stance	Mechanistic claim tested	Key finding	Context	Confidence limitation
	human (genetic)	contradict (direction)	dopamine-receptor efficacy and memory	higher D2/D1/D3 efficacy alleles → better memory		Cognition phenotype

Data and Tool Use

Source / tool	Accession / URI	Access status	Query / usage	Relevant?
PubMed (NCBI E-utilities)	eutils esearch, db=pubmed	Accessed (searched)	13 queries, 2026-09-21 (see <code>search_logs/pubmed_search_log.json</code>)	Yes — primary literature
<code>execute_code</code> sandbox	local	Attempted, failed for artifact writing	blocked <code>hashlib</code> / <code>sys</code> / <code>csv</code> ; <code>PermissionError</code> on <code>kb/</code> path	Tooling only
Bash <code>sha256sum</code>	local	Accessed	checksummed 4 artifact files	Provenance
Datasets / omics / GWAS / cohorts	—	Not accessed	none provided (Files: [])	N/A
Structural files / Phenix	—	Not used	no structures in scope	N/A

Negative searches (index-conditioned, PubMed 2026-09-21): - `aripiprazole tau phosphorylation amyloid Alzheimer model` → **0 results** (central prediction untested). - `dopamine receptor D2 expression Alzheimer disease brain` → no direct DRD2-in-AD-brain quantitation; **DRD2/DRD3 AD-GWAS association is UNVERIFIED** (this index is not a GWAS catalog).

Analysis inventory: Only one "analysis" exists — manual construction of the 22-row evidence matrix from abstract reads (input: PubMed abstracts → method: manual transcription → output: `tables/evidence_matrix.csv`). **Outcome: SUCCEEDED (inspectable CSV).** No statistical/omics analysis was run; those are labeled **SKIPPED**. No prose claim in this report is presented as a computed result.

Mechanistic Causal Chain (upstream → clinical)

1. **D2R over-activation** (*seed premise — CONTESTED in AD; VTA hypodopaminergia and DRD2-A1 genetics point the opposite way:*

P 28367951

P 28965318

P 41930581

2. → **β-arrestin2/PP2A/Akt scaffold, Akt dephosphorylation** (*STRONG as general biology:*

P 17681085

P 23380502

but a distinct arrestin-scaffolded PP2A pool is needed because bulk PP2A is DECREASED in AD:

P 22299660

3. → **GSK-3β Ser9 de-inhibition / activation** (*STRONG edge in principle; but PP2A also dephosphorylates Ser9, muddying sign:*

P 26484916

4. → **tau phosphorylation at GSK-3β epitopes** (*STRONG that GSK-3β→tau:*

P 23321003

P 38160617

but INFERRED for the D2R route specifically)

5. → **neurofibrillary pathology** → **cognitive decline** (*amyloid and PP2A failure are better-supported drivers:*

P 36760217

P 22299660

6. **Drug node:** aripiprazole preserves pAkt-pGSK3 β and antagonizes arrestin (*SUPPORTED off-target:*

P 30597182

P 41192583

→ predicted to **lower** tau; arrestin-biased **UNC9994** predicted to fail/reverse (*MISSING: never measured — the decisive experiment*).

Strong links: steps 2–4 as general signaling biology; the aripiprazole vs haloperidol pAkt-pGSK3 β divergence. **Inferred links:** that step 1 operates in the *pathogenic* direction in AD; that the D2R route (not amyloid/PP2A) sets tau in AD. **Missing steps:** the entire drug→tau readout in an AD/tauopathy model (steps 4–6 for aripiprazole/UNC9994).

Knowledge Gaps

1. **Tau arm never measured (core gap).** Scope: no aripiprazole/UNC9994 study reports tau or pGSK3 β epitopes in any AD/tauopathy model. Checked: PubMed `aripiprazole tau...` = 0 hits (2026-09-21). Resolve: chronic aripiprazole in a tau model (e.g., P301S/rTg4510 or 3xTg) with AT8/PHF-1/pGSK3 β -Tyr216 readouts.

2. **Directional conflict on the dopamine axis (unresolved contradiction).** Seed = D2R over-activation harmful; competing literature = dopamine depletion/low D2 harmful

(P 28367951

P 41930581

P 28965318

). Resolve: dose–response of a D2 partial agonist vs a selective arrestin-biased ligand on tau in the same model.

3. Unconfirmed arrestin-scaffolded-PP2A edge in AD. Bulk PP2A is *decreased* in AD

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P 22299660
),
 so the seed must specify a receptor-proximal PP2A pool. Resolve: D2R/ β arr2/PP2A co-IP and pAkt in AD-model cortical neurons $\pm$ UNC9994.

4. Terminal-node therapeutic doubt. Direct GSK-3 β inhibition failed clinically

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P 31156177
).
 Resolve: confirm target engagement + tau lowering are necessary vs sufficient for cognitive benefit.

5. NILVAD refute is class-level and confounded. Resolve: drug-specific, dose/D2-occupancy-stratified analysis (aripiprazole vs typicals), with neuropsychiatric-symptom adjustment.

6. Source absence (unverified). No DRD2/DRD3 AD-risk association retrievable via this literature index; a GWAS Catalog / GenCC / ClinGen query was **not** executed here and should be logged separately before asserting absence.

Alternative Models

• VTA dopaminergic degeneration / hypodopaminergia

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P 28367951
P 29778899
P 29630556
)
 — *alternative & opposite-direction*: AD is dopamine-deficient; D2 stimulation is protective.

• Amyloid cascade $\rightarrow$ Akt/GSK-3 β /tau

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P 36760217
P 38350636
)
 — *upstream cause*: dominant, D2R-independent driver; the seed's route is a minor tributary.

• PP2A phosphatase failure (APOE- ϵ 4-linked)

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P 22299660
)

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P 28720530)

— *parallel mechanism*: explains tau hyperphosphorylation without D2R.

• **CDK5 as parallel tau kinase**

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P 38350636)

— *parallel mechanism*.

• **c-Abl / autophagy failure**

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P 33684513)

— *upstream/parallel*: drives VTA DA loss; nilotinib rescues.

• **Neuropsychiatric-symptom severity → faster decline (confounding-by-indication)**

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P 34173272)

— *alternative explanation* for the human antipsychotic signal.

Discriminating Tests

1. **Decisive tau experiment.** Chronic aripiprazole vs vehicle vs UNC9994 vs haloperidol in a tauopathy model (P301S or rTg4510). Readouts: AT8/PHF-1/pTau-Ser396, pGSK3 β -Ser9 and -Tyr216, Akt-Thr308/Ser473. *Seed predicts*: aripiprazole ↓ pTau (preserves pGSK3 β -Ser9); UNC9994 no effect or ↑ pTau; haloperidol no protection. *Hypodopaminergia predicts*: benefit tracks restored DA tone, dissociable from arrestin bias.
2. **Dose/occupancy stratification.** Compare 1 mg/kg (5xFAD study) vs therapeutic-occupancy dosing; the seed's arrestin-antagonist reading should be dose-dependent.
3. **Genetic dissection.** β arr2-KO or D2R-neuron-specific GSK3 β -KO crossed to a tau model ± aripiprazole; abolition of effect in β arr2-KO would confirm the arrestin route.
4. **Human observational refinement.** NILVAD/cohort re-analysis: aripiprazole-specific, dose- and APOE-stratified cognitive slopes with CSF p-tau181/217 as biomarker, adjusting for baseline neuropsychiatric symptoms.
5. **iPSC neurons** (APOE ϵ 4 vs ϵ 3, ± tau mutation): aripiprazole vs UNC9994 on pAkt/pGSK3 β /pTau to test the arrestin-scaffolded-PP2A edge in a human, AD-relevant background.

Curation Leads (*require curator verification*)

- **Verify existing snippets:** all five seed references reproduced verbatim in this run

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P 30597182

P 41192583

P 39174788

P 34173272

confirmed; note 41192583 abstract text was not re-retrieved by this index and should be re-checked).

- **Candidate new evidence nodes (competing):**

P 28367951

P 29778899

(hypodopaminergia);

P 22299660

P 28720530

(PP2A/APOE);

P 36760217

(A β →GSK3 β);

P 31156177

(GSK-3 inhibitor RCT null);

P 41930581

(opposite dopamine direction);

P 28965318

P 27647283

(DRD2 genetics).

- **Candidate reagent node (support):**

P 22025698

P 22845053

P 26129680

(UNC9994 = arrestin-biased aripiprazole analogue; discriminating-test tool).

- **Candidate pathophysiology edges to add as HYPOTHETICAL/UNCONFIRMED:**

D2R→ β arr2/PP2A→Akt→GSK3 β →tau (mark tau terminus as unmeasured for aripiprazole in AD); competing edges VTA_DA_loss→hippocampal_hypodopaminergia→memory_deficit and APOE ϵ 4→PP2A_down→tau_P.

- **Candidate ontology terms:** ventral tegmental area dopaminergic neuron (CL:0000700 / UBERON:0001947), GSK-3 β (GO:0016301 tau-protein kinase activity, GO:0004674), β -arrestin-mediated signaling (GO:0002031), protein phosphatase 2A (GO:0000159), tau protein phosphorylation.

- **Candidate status:** keep **EMERGING**; add **knowledge_gaps** for (a) untested tau arm, (b) dopamine-direction contradiction, (c) GSK-3 β -inhibitor clinical null, (d) unverified DRD2 AD-GWAS absence.

Limitations

Literature-only run; no dataset/omics/structural analysis was performed (none provided). The **search_pubmed** index showed low recall on specific queries, so negative searches are **index-conditioned**, not definitive; GWAS/GenCC/ClinGen were not queried. Abstract-level evidence cannot establish effect sizes for the mechanistic Western-blot studies. Artifact bundle: **kb/hypotheses/Alzheimer_Disease/d2r_arrestin_gsk3b_tau_model/openscientist_artifacts/** (MANIFEST.yaml + search log + evidence_matrix.csv + environment + build provenance, sha256-checksummed).